

Controls item table

No.	Type	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote
CTL-1	Control Computer	Control computer	Assembly	Execute operator commands, coordinate subsystems, and manage the fabrication process.	Industrial PC, CPU board, storage, I/O cards, enclosure, power input.	FR-4 PCB, copper traces, aluminum/steel enclosure, copper wiring.	NASA EBF ³ patent describing the instrumentation subsystem. Page p.11. Quote: "computer system."
CTL-2	Control Hardware	Real-time control processor	Assembly	Run deterministic control loops for high-voltage, motion, wire feed, and interlock timing.	Real-time controller board, FPGA/DSP/MCU, memory, clock, isolated I/O.	FR-4 PCB, copper traces, silicon ICs, aluminum heat spreader.	60kV 15kW High voltage tank-BINP Novosibirsk describes DSP and PLM control. Page p.2. Quote: "DSP and PLM."
CTL-3	Control Software	Control software	Software	Command deposition, coordinate beam power, wire feed, motion, data logging, and interlocks.	Path execution module, subsystem command module, safety state machine, recipe manager.	Not a material part; stored on controller storage.	NASA EBF ³ patent describes control software making commands to subsystems. Page p.11. Quote: "control software."
CTL-4	Communication	CAN-bus communication interface	Assembly	Connect high-voltage source, magnetic system, modulator controller, alarm system, and control computer.	CAN interface card, transceiver, isolation, connectors, shield termination.	FR-4 PCB, copper traces, silicon transceivers, shielded twisted cable.	60keV 30kW electron beam-BINP Novosibirsk describes CAN-bus control. Page p.3. Quote: "CAN-bus interface."
CTL-5	Data Acquisition	Analog input and ADC module	Assembly	Measure voltages, currents, temperatures, and feedback signals for control and protection.	Analog input board, ADC, input buffers, isolation, terminal block.	FR-4 PCB, copper traces, silicon ADC, opto/digital isolation, tin-plated terminals.	60kV 15kW High voltage tank-BINP Novosibirsk lists measured channels. Page p.3. Quote: "Measured channels."
CTL-6	Sensor Interface	Sensor interface module	Assembly	Interface thermocouples, pressure sensors, voltage sensors, current sensors, position sensors, and wire-feed sensors.	Signal conditioning circuits, terminal blocks, isolation, shield grounding, calibration interface.	FR-4 PCB, copper wiring, tin-plated terminals, aluminum/steel enclosure.	NASA EBF ³ patent lists instrumentation sensors. Page p.11. Quote: "at least one sensor."
CTL-7	Data Logging	Data logging and timebase module	Assembly	Record process data on a common time base for later analysis and process verification.	Storage device, timing clock, data logger software, synchronization interface.	SSD/flash storage, FR-4 PCB, copper traces, aluminum enclosure.	NASA EBF ³ patent describes recording process data. Page p.11. Quote: "recording data."
CTL-8	Process Monitoring	Visible camera monitoring	Assembly	Visually monitor the	Camera, lens, camera mount,	Aluminum camera housing, glass/fused-	NASA EBF ³ patent describes

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		system		active process zone and chamber interior.	lighting interface, image cable, external recorder/interface.	silica optics, copper cable, polymer insulation outside hot/vacuum zone.	cameras for visual recording. Page p.11. Quote: "video cameras."
CTL-9	Process Monitoring	Thermal imaging monitoring system	Assembly	Monitor melt-pool temperature or thermal field for process control.	Thermal camera, lens/window interface, camera mount, data cable, calibration target optional.	Aluminum housing, germanium/ZnSe/fused-silica optics depending on wavelength, copper cable.	NASA EBF ³ presentation shows thermal imaging for closed-loop control. Page p.29. Quote: "Thermal Imaging."
CTL-10	Motion Control	Motion-control interface	Assembly	Send commands to positioning drives and receive position/status feedback.	Motion I/O card, encoder input, step/direction or servo command output, limit-switch input.	FR-4 PCB, copper traces, silicon ICs, shielded cable connectors.	NASA EBF ³ patent states positioning control is accomplished by instrumentation. Page p.10. Quote: "control of each axis."
CTL-11	Wire Feed Control	Wire-feed control interface	Assembly	Command wire feed rate, start/stop feed, and receive wire-feed sensor data.	Motor command output, feed-rate sensor input, encoder input, interlock input, connector.	FR-4 PCB, copper traces, shielded cable, tin-plated terminals.	NASA EBF ³ patent describes wire feeder control parameters. Page p.11. Quote: "wire feeder control parameters."
CTL-12	Safety Logic	Alarm and blocking logic module	Assembly	Manage interlocks, alarm states, emergency stop logic, and shutdown commands.	Safety relay or safety PLC, interlock inputs, relay outputs, status indicators, wiring terminals.	FR-4 PCB, relay contacts, copper wiring, steel/aluminum enclosure.	60keV 30kW electron beam-BINP Novosibirsk connects alarm/blocking to control computer. Page p.3. Quote: "alarm and blocking system."
CTL-13	Control Cabinet	Controls cabinet	Assembly	House the control computer, I/O modules, communication hardware, power terminals, and safety logic.	Cabinet body, back panel, DIN rails, cable ducts, grounding bar, terminal blocks.	Powder-coated steel, 304/316 stainless steel, 5052 aluminum, copper grounding bar, polyamide terminal blocks.	NASA EBF ³ patent describes internal/external instrumentation and control. Page p.11. Quote: "instrumentation."